School of Innovation and Management (SoIM)
Post Graduate Diploma in Management (PGDM)

Course Outline **PGDM (2026-28)** **Trimester – I**

Course Code:	**Course Name:**	**Course Credits:**
TBA	Operational Analytics	3

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Course Coverage:

Operational Analytics integrates operations management concepts with data-driven decision making. The course covers operational data preparation, KPI analytics, exploratory and root-cause analysis, Power BI control towers, demand segmentation and forecasting, inventory analytics, supplier performance, supply-chain risk, production and capacity analytics, process and bottleneck analysis, statistical process control, machine-learning applications for late-delivery prediction and predictive maintenance, logistics analytics, prescriptive optimization, scenario analysis, and Generative/Agentic AI for operations. Students will work with realistic business datasets using Excel, Power BI and Python to convert operational data into managerial actions.

Course Objectives:

The course aims to develop the ability to understand end-to-end operations flows and translate operational problems into analytical questions. Students will learn to prepare and validate operational data; construct and interpret KPIs; apply descriptive, diagnostic, predictive and prescriptive analytics; evaluate demand, inventory, supplier, production, quality, logistics and risk performance; build management dashboards; and communicate evidence-based operational recommendations. The course emphasizes managerial interpretation, process thinking and the use of analytics to improve service, cost, capacity, quality, resilience and decision speed.

Course Learning Outcomes:

After completing the course, student should be able to:

1. Explain the role of operational analytics in converting operations data into managerial decisions across the value chain.
2. Prepare, clean, validate and structure operational datasets using appropriate data-quality rules and transformations.
3. Calculate and interpret operational KPIs related to service, cost, inventory, suppliers, production, capacity, quality and logistics.

4. Apply exploratory analysis, variability measures, Pareto analysis and root-cause reasoning to diagnose operational problems.
5. Build interactive Excel and Power BI dashboards / control towers for operational monitoring and exception management.
6. Apply demand segmentation, forecasting and inventory analytics to support planning and replenishment decisions.
7. Evaluate supplier performance and supply-chain resilience using normalized scorecards, weighted models, lead-time variability and TTR/TTS concepts.
8. Analyze production capacity, process bottlenecks and quality performance using OEE, takt/cycle time, Theory of Constraints and SPC concepts.
9. Apply predictive and prescriptive analytics, including machine learning, optimization and scenario analysis, to operational decisions.
10. Synthesize analytical evidence into clear managerial recommendations and present an end-to-end operational analytics project.

Assessment Tools for Learning Outcome:

Learning Outcome	Assessment Criteria
Explain the role of operational analytics in converting operations data into managerial decisions across the value chain.	IE-1 case-cracking assignment; class discussion; End-Term Examination.
Prepare, clean, validate and structure operational datasets using appropriate data-quality rules and transformations.	IE-2 analytical assignment / mid-term; practical Excel exercises.
Calculate and interpret operational KPIs related to service, cost, inventory, suppliers, production, capacity, quality and logistics.	IE-2 analytical assignment; dashboard and KPI exercises; End-Term Examination.
Apply exploratory analysis, variability measures, Pareto analysis and root-cause reasoning to diagnose operational problems.	IE-1 case analysis; diagnostic exercises; class discussion.
Build interactive Excel and Power BI dashboards / control towers for operational monitoring and exception management.	IE-2 dashboard assignment; practical Power BI / Excel work.
Apply demand segmentation, forecasting and inventory analytics to support planning and	IE-2 analytical assignment; forecasting and inventory exercises.

Learning Outcome	Assessment Criteria
replenishment decisions.	
Evaluate supplier performance and supply-chain resilience using normalized scorecards, weighted models, lead-time variability and TTR/TTS concepts.	IE-2 supplier / risk analytics assignment; case-based discussion.
Analyze production capacity, process bottlenecks and quality performance using OEE, takt/cycle time, Theory of Constraints and SPC concepts.	IE-2 practical exercises; IE-3 project application.
Apply predictive and prescriptive analytics, including machine learning, optimization and scenario analysis, to operational decisions.	IE-3 project presentation; model / scenario analysis; End-Term Examination.
Synthesize analytical evidence into clear managerial recommendations and present an end-to-end operational analytics project.	IE-3 project presentation and viva; End-Term Examination.

Pedagogy:

The course will be delivered through interactive lectures, case discussions, guided demonstrations and hands-on analytics labs. Students will work with realistic operations and supply-chain datasets using Excel, Power BI and Python. Each topic follows a problem-to-decision approach: understand the operational flow, identify the managerial question, prepare the data, calculate or model the relevant metrics, visualize exceptions, interpret the result and recommend action. Caselets, supplier and inventory exercises, control-tower dashboards, forecasting activities, bottleneck analysis and project presentations will be used throughout the course.

Session Plan: 20 sessions of 90 minutes each, totaling 30 Hours for a 3 Credit course.

S. No.	Topic	Pre-reading / Reference	Pedagogy
1	**Introduction to Operational Analytics** • Operations as an Input–Process–Output–Feedback system • Analytics maturity: descriptive to prescriptive	Faculty notes / selected operations case	Interactive lecture and case discussion. Map an end-to-end operating flow and identify where analytics supports managerial decisions.

S. No.	Topic	Pre-reading / Reference	Pedagogy
	• Operational decision architecture		
2	**Operational Data Preparation** • ERP / transactional data • Data-quality dimensions • Cleaning, standardization and validation	Faculty dataset / Excel functions	Hands-on Excel lab using messy operations data. Students classify completeness, validity, consistency, uniqueness and plausibility issues.
3	**Operational KPI Analytics** • Fill rate, OTD, stockout, cycle time, margin • Ratio construction and aggregation	Faculty notes / KPI reference sheet	Calculate KPIs from transaction-level data and discuss numerator/denominator logic. Compare regional operating performance.
4	**Exploratory Data Analysis & Root Cause** • Distribution and variability • SD, CV, correlation • Pareto and RCA	Faculty dataset / selected caselet	EDA lab in Excel. Identify variation, relationships and dominant causes; distinguish correlation from causation.
5	**Power BI Operations Control Tower** • KPI cards, trends and exceptions • Slicers and drill-down • Management dashboard logic	Microsoft Power BI reference / faculty dataset	Build a one-page control tower. Every visual must answer a management question and lead to an action.
6	**Demand Segmentation: ABC–XYZ** • Annual consumption value • Demand variability • Differentiated planning policies	Faculty notes / ABC–XYZ case	Segment SKUs using value and variability. Discuss AX, AZ, CX and CZ inventory/planning strategies. IE-1 case assignment introduced.

S. No.	Topic	Pre-reading / Reference	Pedagogy
IE-1 (20 Marks) – Cracking an Operations Case Study (Assignment)			
7	**Demand Forecasting** • Moving average and exponential smoothing • Forecast error: MAE, MAPE, RMSE, Bias	Faculty forecasting dataset	Forecast demand in Excel, compare baseline methods and connect forecast error to purchasing, production and safety-stock decisions.
8	**Inventory Analytics** • EOQ, Safety Stock and Reorder Point • Inventory position and Days of Cover	Faculty notes / inventory case	Calculate EOQ, safety stock, ROP and inventory exceptions. Discuss service-level versus working-capital trade-offs.
9	**Supplier Analytics** • Lead-time CV, OTD and defect rate • Normalization and weighted scorecards • Supplier resilience	Faculty supplier dataset	Develop a normalized weighted supplier scorecard; rank suppliers and perform weight sensitivity analysis.
10	**Supply Chain Risk Analytics** • Probability–Impact–Exposure • FMEA / RPN • TTR, TTS and resilience gaps	Faculty risk register / selected case	Build a risk register and heatmap; identify resilience gaps and recommend Avoid/Reduce/Transfer/Accept/Recovery actions.
IE-2 (20 Marks) – Analytical Assignment / Mid-Term			
11	**Production & Capacity Analytics** • Design, effective and actual capacity • Utilization, efficiency and capacity cushion	Faculty production dataset	Diagnose capacity and OEE losses across plants/lines. Decide whether to add capacity, improve process or rebalance production.

S. No.	Topic	Pre-reading / Reference	Pedagogy
	• OEE		
12	**Process & Bottleneck Analytics** • Cycle time, takt time, WIP and flow time • Little's Law • Theory of Constraints	Faculty process-flow dataset	Identify the system constraint and model improvements by reducing cycle time or adding parallel resources.
13	**Quality Analytics & Statistical Process Control** • Variation, control limits and special causes • Defect rate and FPY • Pareto analysis	Faculty quality dataset / SPC notes	Calculate 3-sigma control limits, flag special-cause signals and prioritize defect causes using Pareto analysis.
14	**Machine Learning for Late-Delivery Prediction** • Features, target and train/test logic • Classification metrics • Managerial use of prediction	Faculty ML dataset / Python notebook	Build and interpret a simple late-delivery classification model; connect model output to proactive exception management.
15	**Predictive Maintenance Analytics** • Failure signals and maintenance data • Remaining-life / failure-risk concepts • Maintenance prioritization	Faculty machine dataset	Analyze equipment condition indicators and build a risk-based maintenance prioritization approach.
16	**Logistics & Distribution Analytics** • Transport cost and service • Route / carrier	Faculty logistics dataset	Evaluate logistics performance by region/carrier; identify high-cost or low-service lanes and recommend distribution actions.

S. No.	Topic	Pre-reading / Reference	Pedagogy
	performance • Delivery exceptions		
17	**Prescriptive Analytics & Optimization** • Decision variables and constraints • Objective functions • Allocation / capacity optimization	Faculty optimization exercise / Excel Solver	Formulate a simple operations optimization problem and use Excel Solver to recommend an allocation decision.
18	**Scenario & What-If Analytics** • Sensitivity and assumptions • Capacity, demand and cost scenarios • Decision robustness	Faculty scenario model	Develop base/best/worst scenarios and assess how operating decisions change under uncertainty.
19	**Generative & Agentic AI for Operations** • Natural-language analytics • Operational copilots / agents • Governance and human oversight	Selected readings / faculty demo	Demonstrate GenAI/agentic workflows for operational summaries, exception explanations and decision support; discuss governance.
20	**Operations Analytics War Room – Capstone** • Integrate demand, inventory, supplier, production, quality and risk • Executive recommendation	Project dataset / student-selected operations problem	IE-3 Project Presentation and viva. Teams present the end-to-end problem, analytical evidence, dashboard/model and managerial recommendation.
IE-3 (20 Marks) – Operations Analytics Project Presentation & Viva			
END-TERM EXAMINATION (40 Marks)			

Recommended Text Book:

Operations Management: Sustainability and Supply Chain Management, Jay Heizer, Barry Render & Chuck Munson, Pearson.

Reference Text Books/Recourses (Journals, Internet Websites):

1. Supply Chain Management: Strategy, Planning, and Operation, Sunil Chopra, Pearson.
2. Operations Management, Nigel Slack, Alistair Brandon-Jones and Nicola Burgess, Pearson.
3. Business Analytics: Methods, Models, and Decisions, James R. Evans, Pearson.
4. Microsoft Learn – Power BI and Microsoft Fabric learning resources.
5. Selected HBR / Ivey / practitioner case studies on operations, supply chain, analytics and digital transformation.
6. Faculty-provided datasets, Excel models, Power BI exercises and Python notebooks.

Course Evaluation:

Evaluation is a continuous process at SoIM. Students will be assessed through case-based assignments, analytical exercises, project work and an end-term examination. The evaluation scheme for Operational Analytics is as follows:

Areas	Weightage
IE-1 – Cracking an Operations Case Study (Individual/Group Assignment) Students diagnose an operations case, identify process/supply-chain issues and present evidence-based recommendations.	**20%**
IE-2 – Analytical Assignment / Mid-Term Scenario-based operations and supply-chain analytics using Excel / Power BI.	**20%**
IE-3 – Operations Analytics Project Presentation & Viva End-to-end project using operational data, dashboard/model and managerial recommendations.	**20%**
End-Term Examination	**40%**
Total	**100%**

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